

Developing an Offline Mobile Application with Health Condition Care and First Aid Instruction for Appropriateness of Medical Treatment

Alec Francisco C. Garces and Jose Paolo O. Lojo
La Salle Green Hills, alec.garces060589, jose.lojo080009@my.lsggh.edu.ph

Abstract

Lower income families which comprise a majority of the population have limited access to proper healthcare given scarcity of public doctors and high costs of private healthcare. However, first aid knowledge among the public is not common and incorrect treatments can be counterproductive. With these, the researchers developed a mobile application program to provide instructions for health concerns based around the needs of Our Lady of Banneux, a community made by Answering the Cry of the Poor (ANCOP) and Couples for Christ (CFC), using the community's preferred Filipino language. The community is also a partner community of the researchers' school, La Salle Green Hills (LSGH). Before the program was introduced to the respondents, a survey on the community's socio-demographic profile and a medical knowledge test on common health concerns were given. Latter was done to establish the respondents' baseline first aid knowledge. Participants were then divided into control and test groups with their medical knowledge compared. Using a t-test, the findings suggest that the level of medical knowledge between the groups is statistically the same. To see whether any of the respondents' socio-demographic variables has an association with their medical knowledge, a chi-square test of independence was made and it was revealed that, in both groups, there is not enough evidence to conclude that these have an association. The program was then introduced to the test group and a medical appropriateness test was given to both groups. Based on a t-test done on the scores of the groups, it was found that the program led to significantly higher scores for the test group, indicating that it could help in increasing the appropriateness of first aid interventions.

Key Words: access to healthcare; application; first aid; health instruction; medical appropriateness; medical knowledge; poor communities

Introduction

Accessibility to basic medical knowledge is very limited to low income families especially those living in more rural and isolated areas (Bennett, Yoshizu & Hirai, 2017). Moreover, first aid knowledge among the public is not common and incorrect treatments can be counterproductive (Anwar, Green, Norris & Bukhari, 2015). Based on these, the researchers, Alec Francisco Garces and Jose Paolo Lojo, with the help of their research advisers (Ms. Ma. Lilibeth Bilon and Ms. Alce Sentones), their Practical Research 2 teacher (Mr. Rowell Rosales), the head of the Our Lady of Banneux community (Mrs. Fe Flores), and a researcher consultant (Mrs. Eden Aguirra), developed an offline mobile application in the commonly preferred Filipino language to provide instructions for health concerns according to the needs of a poor community.

Background

Access to health care remains a critical concern, worldwide. A report from the World Bank and the World Health Organization (WHO) states that at least half of the world's population do not have access to basic health services. As a result, a big proportion of families each year are pushed further into poverty because of their large out-of-pocket medical spending (Bennett, Yoshizu & Hirai, 2017). This results to a demand for a way to teach the general public on how to provide themselves basic first aid procedures and medical care. Spies, Khalf & Hamam (2017) were able to develop an application for medical emergency use but it was apparently used for trained individuals only.

After various interviews in the community, it was discovered that, when health problems or injuries arise, most rely on home remedies or medicine and medical devices donated to their clinic, e.g., paracetamol, imodium, rehydration salts, nebulizer, etc.. The community only gets in touch with medical professionals when their situation gets dire. With these in mind, the researchers developed an offline mobile application with healthcare and first aid instructions as a possible aid to the lack of health services.

Process

The researchers created an offline smartphone application because of its accessibility. According to one of the researcher's respondents, a printed handbook or guide would probably get lost. However, a smartphone would always be right by a person's side. Some injuries need emergency action in which a portable guide can be useful. In high pressure situations, stress can cloud a person's memory (Rossman, 2010). An application guide may be able to help even trained individuals in case they forget. Telemedicine is also an alternative option but response time is not as immediate and there may be issues with patient privacy. Results with the use of telemedicine have been mixed in terms of efficiency (Ekland, Bowes & Flottorp, 2010).

Our Lady of Banneux is a community created by ANCOP and CFC which aim to provide homes for the poor and the needy. The community was chosen because of its isolated nature. It is distant from the city and from a hospital. In case of an emergency, the program the researchers created could help mitigate injury while waiting for medical help.

The three-part study began with a community profiling where researchers determined the community's common health concerns and socio-demographic profile. The second part involved classifying the respondents into two groups based on smartphone ownership: one group without (control) and another with smartphones (test). This stage established and compared the two groups' baseline first aid knowledge through a medical knowledge test, the content of which was reviewed and checked by practitioners. The control and test groups' medical knowledge was also checked for association with their socio-demographic profile. Development of the application then began with the content guided by the community's health concerns, sourced from the Red Cross First Aid app and Mayo Clinic website, and validated by medical practitioners. The app is programmed to operate offline and gives users first aid instructions based on the symptom / injury that they select. It provides visuals of the symptom / injury and gives instructions in Filipino for easier understanding by the users in the community. With the completed application, the researchers went into the final stage of the study where a medical appropriateness test was given to both the control and test groups. The test which was checked by medical practitioners sought to determine whether the app will lead to significantly better scores for the members of the test group who were given the application.

Results

Result of the t-test for medical knowledge test suggested that, at a 95% confidence level, medical knowledge between the two groups is statistically the same having a t-value of -1.64 within the non-rejection region.

Additionally, the chi-square test of independence revealed that in both groups, there is insufficient evidence to conclude that their level of medical knowledge has an association to their socio-demographic profile at 0.05 level of significance.

Based on a t-test done on the scores of the groups, it was found that the program may have led to higher scores for the test group. The results showed that the t-value of -2.19 is below the critical value for a left-tailed test which indicates that the app may be able to help increase the appropriateness of first aid interventions.

Future Work

Future studies may consider improving the content of the app, e.g., adding more visual aids and causes of injuries, to facilitate understanding. Other languages may also be included to lend help to other users. The app and the results of the study may also be presented to the Philippine Department of Health as a possible assistance to Geographically Isolated and Disadvantaged Areas (GIDA) in the country.